GREEN ALIBI — Development Log

This is my working log for GREEN ALIBI, a project testing whether Solar-Induced Fluorescence (SIF) — a direct physical proxy for photosynthetic activity — can detect crop and vegetation drought stress earlier than the vegetation-greenness index (NDVI) that India's official drought-monitoring framework currently relies on. I kept this log the way I keep every project's log — as an honest, chronological record of the decisions, dead ends, and fixes that went into the analysis, not a cleaned-up summary written after the fact.

What follows is organized as a set of entries, each covering one phase of the work: framing the research question, building the SIF and NDVI extraction pipelines, catching and fixing a boundary-precision problem, bringing rainfall in as an independent validation, adding a second, independent lag-estimation method, a full self-review pass, and finally expanding the study from three years to eight. Every number, correlation, and decision below reflects what I actually found when I ran the analysis, including results that didn't cleanly support my original hypothesis.

Index

1. Entry 1
2. Entry 2
3. Entry 3
4. Entry 4
5. Entry 5
6. Entry 6
7. Entry 7
8. Entry 8
9. Entry 9
10. Entry 10
11. Entry 11
12. Entry 12
13. Entry 13
14. Entry 14
15. Entry 15
16. Entry 16

Entry 1

GREEN ALIBI is an independent satellite-verification study testing whether Solar-Induced Fluorescence (SIF) — a direct physical proxy for actual photosynthetic activity — can detect crop and vegetation drought stress earlier than the vegetation-greenness index (NDVI) that India's official drought-monitoring framework currently relies on. NDVI is a reflectance-based index: it measures how much visible and near-infrared light a leaf's surface reflects, and that reflectance only changes once a plant's internal structure has already begun to visibly degrade — thinning leaves, reduced chlorophyll content, canopy stress. SIF measures something physically earlier and more direct. When a leaf absorbs sunlight for photosynthesis, a small fraction of that absorbed energy is not converted into chemical energy but is instead re-emitted as fluorescence — a process governed by the quantum yield of photosynthesis. When a plant is under physiological stress, its photosynthetic efficiency drops before its outward greenness does, and this shows up as a measurable change in fluorescence emission well before NDVI shows any decline. In effect, NDVI shows a plant's alibi — it can still look green — while SIF shows what is physically happening inside it.

India's drought declaration process, and the crop-insurance payout mechanism under the Pradhan Mantri Fasal Bima Yojana (PMFBY) that depends on it, is built substantially on rainfall-deficit and NDVI-based assessment. Both are reflectance- or precipitation-based indicators, and both are known to respond only after visible crop stress has already set in. Delayed drought recognition means delayed insurance payouts, at exactly the point when affected farmers need them fastest. This project treats that delay as an empirical, testable question rather than an assumption: if SIF captures physiological stress earlier than NDVI, and if that lag is large and consistent enough to matter, it represents a concrete, physics-based case for an earlier-warning indicator than the one currently used in policy.

My aim was to independently test, using satellite-derived SIF and NDVI datasets over drought-affected agricultural districts in India, whether SIF shows a measurably earlier decline than NDVI during known drought episodes, and whether the resulting lag is large enough and consistent enough to carry practical early-warning value for drought declaration and crop-insurance timelines. That aim broke down into four research questions: whether SIF declines measurably earlier than NDVI during documented drought-onset periods in the study region (RQ1); whether the SIF–NDVI lag, where it exists, is consistent across different districts and crop types, or varies meaningfully by geography and crop physiology (RQ2); how the timing of SIF-based stress onset compares against official state drought-declaration dates for the same districts and years (RQ3); and whether the observed SIF–NDVI lag translates into a practically meaningful early-warning window — measured in weeks — relevant to real crop-insurance and drought-relief timelines (RQ4).

Going in, I held three hypotheses: that SIF would show a statistically significant decline before NDVI during the same drought episode, consistent with fluorescence capturing a drop in photosynthetic efficiency ahead of any visible, reflectance-based greenness loss (H1); that the magnitude and timing of SIF–NDVI divergence would not be uniform across crop types, since fluorescence response depends on canopy structure and species-specific photosynthetic pathways (H2); and that official drought declarations would lag behind the SIF-based stress signal, consistent with the official process's reliance on slower-responding rainfall and NDVI indicators rather than a direct physiological measure (H3).

At this earliest planning stage, the foundational tasks ahead were: selecting a study region with a well-documented recent drought history and reliable official declaration records — likely candidates included the Marathwada and Vidarbha regions of Maharashtra and the Bundelkhand region spanning Uttar Pradesh and Madhya Pradesh, all of which have recurring, well-reported drought episodes; sourcing a usable SIF dataset, most likely the gridded GOSIF product or a TROPOMI-derived SIF proxy, since native fine-resolution SIF instruments are not designed for agricultural-field-scale monitoring; sourcing matching NDVI data from MODIS or Sentinel-2 for the same districts and time windows; and identifying accessible official drought-declaration records and, where possible, PMFBY payout timing data to serve as the real-world policy benchmark against which the satellite-derived signal would be compared. None of this data had been acquired yet at this point — this entry marks the start of that process, not its outcome.

Entry 2

With the project framed in Entry 1, the next task was building the actual satellite pipeline — sourcing SIF data, sourcing a matching NDVI dataset, and producing the first real comparison between the two for Marathwada.

Study region and years finalized. Marathwada was chosen as the study region — a personal choice as much as a methodological one, since it is my own home region and has a well-documented recent drought history. Three years were selected for the season of June 1 to November 1 (day-of-year 153–305, the Kharif growing season): 2015 and 2018, both documented Marathwada drought years (2015 in particular being the year of the severe Latur water crisis), and 2020, a comparatively normal monsoon year, as a baseline for comparison.

Sourcing SIF data. GOSIF v2 — a global, 0.05-degree, 8-day solar-induced fluorescence product derived from OCO-2, MODIS, and reanalysis data (Li & Xiao, 2019), distributed by the Global Ecology Group at the University of New Hampshire — was identified as the most usable SIF source, since native fine-resolution SIF instruments are not designed for field-scale agricultural monitoring. Sixty 8-day GeoTIFF files were downloaded in total (20 each for 2015, 2018, and 2020, covering the same day-of-year range).

Clipping and scaling. A script was written to clip each global GeoTIFF to a Marathwada bounding box (17.5°N–20.5°N, 75.0°E–78.5°E, covering all eight Marathwada districts), and, critically, to correctly handle GOSIF's raw digital values: a scale factor of 0.0001 needed to be applied to convert raw pixel values into actual SIF units, and two fill-value codes (32766 for water bodies, 32767 for non-vegetated or missing data) needed to be masked out before any averaging. Skipping this step would have silently corrupted every downstream mean, since the fill values are numerically far outside the real SIF range and would have dominated any naive average. A second script then aggregated each clipped file into a single Marathwada-wide mean SIF value per date, also recording the fraction of valid (non-masked) pixels per date as a data-quality check.

A serious cloud-masking bug in the NDVI extraction. NDVI was sourced separately via Google Earth Engine, computed from MOD09Q1 (MODIS 8-day surface reflectance) so that its temporal resolution would match GOSIF's 8-day cadence exactly. The first version of this script computed NDVI directly from raw reflectance bands without applying any cloud-quality filtering. The resulting NDVI time series was implausibly noisy — during the 2020 season in particular, NDVI swung from roughly 0.55 down to 0.14 and back up to 0.60 within consecutive 8-day periods, a physically impossible rate of change for cropland vegetation. Comparing this against the SIF series, which was smooth and phenologically sensible, made clear that the NDVI series itself was the problem, not the underlying vegetation signal. The cause was identified as cloud and cloud-shadow contamination: Marathwada's growing season coincides with the monsoon, and MOD09Q1's raw reflectance, without explicit quality-band filtering, does not reliably exclude cloud-affected pixels from its "best observation per 8-day window" selection.

Fix: switching to MOD13Q1 with explicit quality masking. The NDVI extraction was rebuilt around MOD13Q1, NASA's own 16-day Vegetation Index product, which applies a Maximum-Value-Composite algorithm specifically to suppress cloud contamination, and which ships with a `SummaryQA` band (0 = good, 1 = marginal, 2 = snow/ice, 3 = cloudy) that was used to explicitly mask out anything but good-to-marginal-quality pixels. A cropland mask (MCD12Q1 land-cover product, IGBP classes 12 and 14) was also applied at this stage, so that forest, urban, and barren pixels within the bounding box would not dilute a signal that is specifically about crop stress. This traded 8-day temporal resolution for 16-day, but produced a visibly smooth, physically sensible NDVI curve — a trade worth making, since a clean 16-day signal is more trustworthy than a noisy 8-day one.

A second, smaller bug: an Earth Engine clip error. The corrected script initially failed with `Image.clip: Can't transform (0.0,0.0)`, caused by calling `.clip()` on an image built from bands of two different native resolutions (250m NDVI, 500m-derived cropland mask). The fix was to remove the `.clip()` call entirely, since `reduceRegion()` already restricts its calculation to the supplied geometry — the clip step was redundant and was the one throwing the error.

Merging two time series at different temporal resolutions. Since SIF (8-day) and NDVI (16-day, after the fix) no longer shared an identical date grid, the two series were joined using nearest-date matching per year, rather than an exact date join, so that each 8-day SIF value was paired with its closest available 16-day NDVI value.

A labelling bug caught before it could mislead. An early version of the comparison plot labelled 2018 as a “normal monsoon year,” when it is in fact the second of the two documented drought years in this study. This was a conditional-logic oversight (only 2015 had been explicitly flagged as a drought year in the plotting code) rather than a data error, and was corrected before any interpretation was drawn from the mislabeled chart.

First seasonal comparison, and an honest correction of my own overclaim. With the corrected pipeline, SIF and NDVI trajectories were plotted for all three years. Visually, a consistent pattern holds across all three years: SIF begins declining from its seasonal peak before NDVI does, with NDVI holding near its peak value for a noticeably longer period after SIF has already started dropping. This is a genuine and replicated pattern, and it directly supports the project’s core premise — that SIF is more responsive to the onset of post-peak physiological change than NDVI is. However, an initial, more casual visual read of the three charts led me to claim that this lag was distinctly larger in the two drought years (2015, 2018) than in the normal year (2020). Checking that claim properly, by computing each year’s decline as a percentage of its own peak value at each subsequent date rather than just eyeballing the chart, did not support it: the apparent lag was of a broadly similar magnitude (roughly 15–25 days) across all three years, drought and normal alike, once measured carefully. That earlier claim is retracted here rather than left standing — the correct, defensible finding at this stage is that SIF leads NDVI’s decline in general, not that drought years show a demonstrably larger lag than normal years. Establishing whether drought specifically amplifies this lag will require a proper quantitative lag metric (a day-of-year-to-threshold comparison or a formal cross-correlation calculation) rather than visual comparison, and is the next task.

Status after this entry. The SIF and NDVI extraction pipelines are both built, cloud/quality-screened, and cropland-masked, covering three years (2015, 2018, 2020) over Marathwada. A real, replicated SIF-leads-NDVI pattern has been observed. What remains before this can be reported as a finding is a rigorous, numeric lag calculation in place of the visual comparison used so far, and — ideally — at least one additional comparison year to strengthen the small sample this analysis currently rests on.

Entry 3

Entry 2 ended with an unresolved problem: the first quantitative lag calculation (comparing how many days after its own peak SIF and NDVI each crossed a series of decline thresholds) produced results that could not be trusted, because the observation window — June 1 to November 1 — ended before NDVI had finished declining in two of the three years. Several thresholds simply never resolved (returned no crossing date at all), and the ones that did resolve were being averaged unevenly across years, which made any comparison between drought and normal years unreliable rather than simply inconclusive.

Extending the window. The fix was to extend data collection further into the year, adding seven more 8-day GOSIF periods (day-of-year 313, 321, 329, 337, 345, 353, and 361) for all three years, pushing the observation window from November 1 out to late December. The same clipping and aggregation scripts built in Entry 2 picked up these additional 21 files without any code changes, since both were written to scan for whatever GOSIF files exist in the raw data folder rather than assuming a fixed date range. The NDVI extraction script's end date was likewise extended from November 1 to December 31, and the same MOD13Q1 cloud-screened, cropland-masked pipeline was re-run for the longer window.

The extension worked as intended. With the longer window, both the SIF and NDVI seasonal curves now taper down smoothly through day 350-360 in all three years, with no sign of the noise or truncation that motivated this fix. More importantly, every single threshold (90%, 80%, 70%, 60%, and 50% of seasonal peak) now resolves to an actual crossing date for all three years — the earlier problem of unresolved, missing thresholds is gone, and the year-to-year comparison can now be made on genuinely equal footing.

The result, once the comparison is fair, does not support the hypothesis as originally framed. Mean lag across all five thresholds came out to 21.6 days for 2015 (drought), 8.4 days for 2018 (drought), and 28.5 days for 2020 (the normal-monsoon comparison year). Averaged by group, the two drought years show a smaller mean lag (15.0 days) than the single normal year (28.5 days) — the opposite of what H1 predicted. This is not an artifact of the earlier truncation problem: with the extended window, 2020's lag at the 70%, 60%, and 50% thresholds forms a smooth, consistently large curve (36.9, 38.9, and 39.5 days respectively) rather than a single anomalous spike, which is what would be expected if this were still a data-window artifact. The result appears to be genuine.

An additional complication: the two drought years do not behave alike. 2015's mean lag (21.6 days) is roughly two and a half times 2018's (8.4 days), despite both being documented Marathwada drought years. This suggests that "drought year" as a simple binary label may be too coarse a category for this analysis — the two droughts likely differed in onset timing, severity, or duration in ways that matter for this specific comparison, and a sample of two drought years is nowhere near enough to characterize that variation, let alone average over it meaningfully.

What does still hold up. Across all three years — drought and normal alike — SIF consistently begins its post-peak decline before NDVI does, with NDVI holding near its peak value for a longer stretch after SIF has already started dropping. This part of the finding is robust and replicated three times over, and it supports the project's core premise: SIF is a more temporally responsive proxy for the onset of physiological decline than NDVI is. What the data does not support, at least not with this sample, is the more specific claim that drought stress amplifies this lag relative to a normal season. If anything, the limited evidence available points the other way, though with only three years (two drought, one normal) this is far too small a sample to treat as a confident finding in either direction — it is a genuine, honestly-reported non-result on the drought-amplification question, not a confirmed contrary finding.

Status after this entry. The SIF-leads-NDVI decline pattern is now a solid, three-times replicated observation for Marathwada, extending across the full growing season through December. The specific hypothesis that drought years show a larger SIF-NDVI lag than normal years is not supported by this data, and the two drought years studied differ from each other by more than either differs from the normal year, which complicates any binary drought/non-drought framing of this particular comparison. This is being reported as it stands rather than adjusted to fit the original hypothesis — the honest result here is a replicated SIF-leads-NDVI pattern with an inconclusive, and possibly contrary, relationship to drought severity specifically, at this sample size.

Entry 4

After I had a complete, extended-window lag analysis behind me (June–December, 2015/2018/2020, all fifteen threshold-year combinations resolved), I moved on to what I'd been putting off: actually looking at SIF spatially instead of only as a whole-region average. I built a first script, `map_sif_spatial.py`, to plot the raw clipped SIF rasters for a single day (day-of-year 273, right around the point where the seasonal decline is well underway) side by side for all three years, on a shared colour scale.

The first version of that map genuinely looked good — a visibly concentrated low-SIF (red) patch sitting in the west-southwest part of my study area in 2015 and 2018, and a much greener 2020 panel. That's roughly where I'd expect stress to show up if the underlying idea — that SIF flags drought earlier and more precisely than NDVI — has any spatial coherence to it, not just a temporal one.

But looking at that map made me actually stop and ask a question I should have asked much earlier: what is my study area, exactly? Up to this point, every single script in this project — the GOSIF clipping, the NDVI extraction in Earth Engine, all of it — had been built around one rectangle: 75.0–78.5°E, 17.5–20.5°N. I picked those numbers early on as "roughly Marathwada" and never went back to check them against anything more precise than my own eyeballing of a map.

So I actually opened Earth Engine's Code Editor and added the two lines I'd never bothered to add before — `Map.centerObject()` and `Map.addLayer()` — just to draw that rectangle and actually look at it zoomed in, instead of trusting it blindly. And once I did, the problem was obvious: the box, while covering all eight Marathwada districts, also ate a meaningful chunk of Telangana (parts of Nizamabad, Bidar, and Adilabad districts), touched Solapur, and clipped a corner of Yavatmal. None of those are Marathwada. They also don't share Marathwada's drought history — Solapur and the Telangana districts sit in different rainfall regimes entirely. Every mean SIF and mean NDVI value I had computed up to that point had quietly been an average over Marathwada-plus-a-noticeable-slice-of-somewhere-else, not over Marathwada alone.

I decided this wasn't something I could let go, given how central "this is a Marathwada-specific finding" is to the whole point of this project. A rectangle can never actually match Marathwada's real outline anyway — the region has an irregular shape, with Nanded pushing east and Osmanabad pushing south — so the right fix wasn't to fiddle with the rectangle's corners, it was to stop using a rectangle at all.

I switched to Earth Engine's FAO GAUL 2015 level-2 administrative boundaries dataset, filtered it to Maharashtra state and to the eight district names that make up Marathwada, and merged them into a single precise polygon to replace the rectangle everywhere in the NDVI extraction script.

The first time I ran this, I immediately found a bug in my own fix: when I drew the new polygon, there was a visible gap right where Beed district should have been — the rest of Marathwada filled in correctly, but Beed was just missing. I'd filtered on the district name `'Beed'`, which is the name almost everyone uses today, but the FAO GAUL 2015 dataset — being older — stores this district under its earlier spelling, `'Bid'`. Since my filter string didn't match anything in the dataset, that one district silently dropped out of the geometry, and nothing warned me about it — the script ran without error, it just quietly returned seven districts instead of eight. I only caught it because I happened to look at the map instead of trusting the script's silence. I fixed the name and added an explicit count check (`.size()`) after the filter, printing the number of matched districts, so if this ever happens again with any dataset naming quirk, I'll get a number that's wrong instead of a shape that's wrong and easy to miss.

With the NDVI side fixed and confirmed against an academic reference figure of Marathwada's district boundaries I found online, I still had the SIF side to correct. The GOSIF clipping script had been using `rasterio`'s window-based cropping, which is inherently rectangular — it has no way to represent an irregular polygon boundary. To fix it properly, I exported the same precise Marathwada polygon out of Earth Engine as a GeoJSON, brought it into my local project, and rewrote `clip_gosif.py` to use `rasterio.mask.mask()` against that polygon instead of a bounding-box window. Every pixel that falls outside the real Marathwada district outlines — including everything that used to leak in from Telangana, Solapur, and Yavatmal — is now set to NaN rather than being included in the regional mean.

I re-ran the full chain after that: `clip_gosif.py`, then `aggregate_sif.py`, then `compare_sif_ndvi.py`, then `lag_analysis.py` — all four in sequence, on the corrected boundary, for both datasets.

The result I was bracing for was that the numbers might shift and possibly change my conclusion. That's not what happened. The sanity-check SIF means by year (2015: 0.137, 2018: 0.151, 2020: 0.219) preserved the same drought-vs-normal ordering as before. The seasonal SIF-vs-NDVI trajectories still showed SIF's decline consistently preceding NDVI's decline in all three years. And the lag numbers, while not identical to the earlier (contaminated) run, told the same story: mean lag across thresholds was 24.2 days for 2015, 5.1 days for 2018, and 25.5 days for 2020 — drought-year average (14.6 days) still smaller than the normal year's (25.5 days), so the hypothesis that drought amplifies the SIF-to-NDVI lag is, again, not supported. If anything, the gap between my two drought years (2015 and 2018) came out even more pronounced with the cleaner data than it had with the contaminated rectangle.

I'm treating that as a genuinely useful result rather than a disappointing one. Getting the same qualitative answer from two different, independently-built definitions of the study region — a rough rectangle and a precise administrative boundary — is a real check on whether the earlier finding was actually a signal or just an artifact of sloppy clipping. It held up. That doesn't make the drought-amplification hypothesis correct, but it does mean I can now say with more confidence that its rejection isn't a mistake caused by including the wrong pixels.

What I still don't have: any ground-truth confirmation that the specific low-SIF patch I saw in the spatial map genuinely lines up with cropland rather than some other land-cover class sitting inside the same pixels, and the sample is still only three years, which limits how far I can generalize any of this. Both of those are honest limitations I'm carrying into the next stage rather than papering over.

Next step: regenerate the spatial SIF map using the now-corrected clipped rasters (the underlying files were already overwritten by the boundary fix, so this should just be a re-run, not a rewrite), and then move on to the interactive Folium version.

Entry 5

With the boundary-precision correction from the previous entry done — GOSIF re-clipped with the real eight-district polygon instead of a rectangle, NDVI re-extracted the same way — the obvious next step was to regenerate the spatial comparison map for day-of-year 273 across 2015, 2018, and 2020, this time expecting it to visibly show Marathwada's actual irregular outline instead of a plain box.

It didn't. When I reran `map_sif_spatial.py`, the output still looked like a smooth, fully-coloured rectangle, no different in shape from the version I'd made before the fix. That was worrying in a specific way: either the boundary correction hadn't actually taken effect on the raster data, or it had taken effect but the plotting script wasn't showing it. Those are two very different problems, and I didn't want to guess which one I was looking at.

So instead of trusting either a good-looking or a bad-looking picture, I built a small, separate diagnostic script whose only job was to answer that question directly: it opened one clipped SIF file, plotted it, and drew the actual Marathwada boundary polygon as a blue outline directly on top of the same axes, in the same coordinate space. If the coloured data extended past that blue line anywhere, the masking had failed. If it stopped exactly at the line, the masking was correct and the earlier "looks like a rectangle" impression was something else.

The overlay came back clean — the SIF data stopped exactly at the boundary line everywhere, including at the smaller notches and the eastward hook near Nanded. So the clipping fix from the previous entry was genuinely working. That meant the problem was in the map script, not the data.

Looking at `map_sif_spatial.py` again, the reason became obvious: I'd fixed `clip_gosif.py` to mask against the real polygon, but I'd never gone back and updated the multi-panel comparison script to match. It was still displaying each raster using a fixed rectangular extent left over from before the boundary fix, and it never drew the boundary outline at all — so even though the underlying array had the correct NaN pattern outside the real district shapes, nothing in the plotting code made that visible at this scale across three side-by-side panels.

I rewrote the script to pull each raster's own true bounds directly from its file (rather than a hardcoded rectangle) and to draw the same boundary polygon outline on every panel, on top of the shared colour scale and the panel-spacing fix from earlier. I also renamed the output file with a version suffix rather than overwriting the old one silently, the same habit I've been using for the Google Drive exports, so there's no ambiguity later about which image is the corrected one.

The regenerated figure now shows what it should: all three panels display Marathwada's real, irregular outline, data confined exactly within it. The pattern itself held up well under this more careful rendering — 2015 and 2018 both show a distinct, spatially coherent low-SIF zone concentrated in the western edge of Aurangabad district running down through the Latur–Osmanabad belt, while 2020 is overwhelmingly high-SIF (green) across almost the entire region, with only a small patch of lower values in the south.

The lesson I'm taking from this one is less about the map itself and more about process: a figure looking visually plausible isn't the same as it being verified. I could have looked at the first regenerated (still-rectangular) version, decided it was "probably fine since the underlying pipeline was fixed," and moved on — the picture wasn't obviously wrong, it just wasn't showing what it should have been showing. Building the standalone overlay check instead of trusting my own read of the image is what actually caught that the map script itself was the remaining stale piece.

What's still outstanding: this is a single day-of-year snapshot (DOY 273) rather than a full seasonal animation, and the low-SIF zone I'm seeing still hasn't been checked against any actual ground-level drought impact reporting for these districts — it's consistent with what I'd expect from known Marathwada drought geography, but "consistent with expectation" is not the same as "confirmed." Both are worth flagging honestly rather than treating this map as more conclusive than it is.

Entry 6

Up to this point, "2015 and 2018 were drought years, 2020 was a normal monsoon year" had been a label I was carrying in from general knowledge about Marathwada, not something I had actually measured within this project. Before moving on to writing anything up, I wanted to close that gap, so I brought in CHIRPS daily precipitation data through Earth Engine — the same region, the same June-to-December window I'd been using for SIF and NDVI — and, alongside the three study years, pulled a twenty-year (2001–2020) climatology for the same window so I'd have an actual normal to compare against, rather than comparing the three years only to each other.

The result was a genuine, useful validation: the twenty-year mean was 826.4 mm for the season, with a standard deviation of 158.8 mm. Against that, 2015 came in at 648.9 mm (−21.5% below normal), 2018 at 674.8 mm (−18.3% below normal), and 2020 at 1066.6 mm (+29.1% above normal). So the drought/normal labels I'd been using throughout this project are now backed by measured precipitation deficit, not assumption — both 2015 and 2018 were real, substantial rainfall shortfalls, and 2020 was a genuinely wet year by comparison.

But putting this number next to the SIF-NDVI lag results from earlier made something else visible that I hadn't expected, and I think it's worth recording honestly rather than leaving out because it doesn't simplify the story. 2015's rainfall deficit (−21.5%) and 2018's (−18.3%) are close to each other — a difference of only about three percentage points. But the mean SIF-to-NDVI lag in those same two years was 24.2 days in 2015 versus just 5.1 days in 2018 — nowhere close to each other. If the size of the rainfall deficit were what drove the size of the lag, I’d expect these two similarly-dry years to produce a similarly-sized lag, and they don’t. Whatever is producing the very different lag behaviour between these two drought years, it isn’t simply explained by how much rainfall was missing that season.

I’m treating this as an honest limitation to state plainly rather than something to explain away. It’s possible the answer is about timing within the season (a deficit concentrated early versus late in the monsoon could affect the SIF-NDVI relationship very differently even at the same total shortfall), or about the starting soil moisture condition heading into each season, or about something else this dataset doesn’t capture at all. I don’t have the data in this project to test any of those explanations yet, and I’m not going to guess at one just to make the story tidier.

What this rainfall addition does give me, concretely: the drought-year framing that runs through this entire project is no longer just inherited knowledge, it’s independently measured within the same pipeline as everything else. And the finding that “drought amplifies the SIF-NDVI lag” is not supported still stands, now on a firmer footing — with actual rainfall data in hand, it’s clear this isn’t a case of one of my “drought years” secretly not being that dry after all.

Entry 7

With the district-level rainfall anomaly numbers already validated at the regional level, I wanted the same kind of spatial view for rainfall that I’d already built for SIF — not just a single number per district printed in a table, but something visual, and something that let me directly compare where the rainfall deficit sat against where the SIF stress showed up.

I built this in three layers, matching the pattern I’d already established for SIF. First, an interactive Folium choropleth — each of the eight districts colored by its rainfall anomaly percentage, with a layer for each of the three years, hoverable and clickable, matching the same style as the SIF-by-district Folium map. Second, a static version of the same thing using GeoPandas and matplotlib — three panels side by side, one per year, with each district’s exact anomaly percentage labeled directly on the map, meant for the paper rather than the browser.

The two on their own already told a clear story. In 2015, the western and southern districts — Aurangabad, Bid, Jalna, Latur, and especially Osmanabad — all showed rainfall well below normal, while Hingoli and Nanded in the east were only mildly below normal. In 2018, that same west-to-east pattern showed up again, but sharper: Aurangabad and Bid were both more than 35% below normal, while Hingoli and Nanded actually had a rainfall surplus that year. That’s worth sitting with for a second — the region-wide number I’d computed earlier called 2018 an 18% deficit year, and that’s true on average, but it hides that the deficit was almost entirely concentrated in the west and south, while the east was doing fine or better. “Drought year” as a single label for the whole region flattens something that was actually quite unevenly distributed on the ground.

The third piece was the one I actually wanted from the start: a combined figure with SIF on top and rainfall anomaly underneath, same three years, side by side, so the two could be read against each other directly instead of me trying to hold both maps in my head at once. Putting them together made the spatial correspondence obvious in a way the separate maps didn’t quite manage — the deep red, low-SIF zone in the southwest in 2015 and 2018 sits almost exactly on top of the deep red, rainfall-deficit zone in the same two maps, and the same districts that turned green with rainfall surplus in 2020 are the same ones that turned green with healthy SIF that year. Nanded, which barely dipped in rainfall in either drought year, is also the district that stayed comparatively green in SIF in both of those years. That's a genuinely reassuring, physically coherent picture — two independently measured satellite products (a fluorescence signal and a precipitation product, from entirely different sensors) pointing at the same places for the same reasons.

Building the combined figure surfaced a small but real presentation bug: the overall figure title and the six individual panel titles were rendering too close together, visually merging into each other. I fixed the spacing by giving the subplot grid more headroom (reducing how much of the figure height the panels themselves occupy) and adding explicit padding to each panel's own title, rather than just nudging the suptitle position by itself — the two titles need separated space, not just separated coordinates.

What I'm not claiming from this: visual correspondence between two maps, however striking, is not a statistical test. I haven't run any actual correlation or regression between district-level rainfall anomaly and district-level SIF across the three years — with only three years and eight districts, that would be a weak analysis anyway (an n of 24 district-year pairs, but not independent since districts within a year are spatially correlated with each other). What I have is a visual, physically-motivated consistency check, and I'm describing it as exactly that and nothing stronger.

Entry 8

Entry 7 ended with a deliberate limitation: I had a combined SIF-and-rainfall map that looked convincing side by side, but I explicitly held back from calling it anything more than a visual, physically-motivated consistency check, because I hadn't actually run a correlation between the two. Before treating this project as finished, I went back and did exactly that — the one thing I'd flagged as missing rather than left silently undone.

I merged the two datasets I already had sitting in `data/processed/` — `sif_by_district.csv` and `rainfall_anomaly_by_district.csv` — on district and year, giving 24 district-year pairs (8 districts × 3 years). Against those 24 pairs, I ran both a Pearson correlation (assumes a linear relationship) and a Spearman correlation (rank-based, doesn't assume linearity) between mean SIF and rainfall anomaly percentage, rather than relying on just one.

The result came back stronger than I expected: Pearson $r = 0.837$ ($p < 0.001$), Spearman $\rho = 0.857$ ($p < 0.001$). Both measures agree, and agree strongly — this isn't a borderline or ambiguous number sitting near significance, it's a clear, high-confidence positive correlation between rainfall deficit and SIF suppression across the districts and years I studied.

I'm not treating this as license to overclaim, though, and the same caveat I wrote in Entry 7 about independence still applies, just more precisely now. Twenty-four data points sounds like a reasonable sample size, but the eight districts within any given year aren't independent of each other — they're geographically adjacent and share the same regional weather systems, so the real number of independent observations here is closer to three (one per study year) than twenty-four. I'm reporting the correlation exactly as calculated, without adjusting for this, but I'm also not pretending 24 independent data points went into it when they didn't.

What this changes: the "visual, physically-motivated consistency check" framing from Entry 7 is now backed by an actual number, and a strong one. What it doesn't change: the core finding from Entries 3, 4, and 6 — that drought severity doesn't cleanly amplify the SIF-NDVI lag — is untouched by this. This correlation is about spatial agreement between SIF and rainfall, a separate question from the temporal lag question, and I don't want the two to blur together just because both turned out to support the project's underlying premise that SIF and rainfall are measuring real, physically connected things.

Entry 9

Going back through my own Physical Basis writeup for the research paper, I noticed I'd stated a fact about GOSIF without following it to its actual implication. I'd written that GOSIF "is not itself a direct satellite retrieval, but a statistically modeled reconstruction combining discrete OCO-2 SIF soundings with continuous MODIS reflectance data and meteorological reanalysis." That's accurate — it's how the product is built. What I hadn't spelled out anywhere is what that means for this specific project: the same MODIS reflectance data that GOSIF uses to interpolate SIF values between OCO-2 overpasses is also the underlying data source for the NDVI I'm comparing it against.

In practice, this means SIF and NDVI in this study are not built from two fully independent measurements. On dates and locations without a direct OCO-2 sounding, GOSIF's estimate is partly informed by the same reflectance patterns that produce the NDVI value at that same pixel. This doesn't mean the SIF-leads-NDVI lag finding is wrong, and it doesn't mean GOSIF is a bad dataset to use — it's the standard, widely-used continuous SIF product for exactly this reason, since raw OCO-2 soundings alone are too sparse in space and time for a regional study like this one. But I hadn't written down, anywhere in this project, the specific consequence that matters here: some degree of shared information between my two variables is baked into the dataset choice itself, and I have no way, with the data and time available to me, to quantify how much of the observed lag is attributable to genuine independent fluorescence physics versus this modeling dependency.

I'm documenting this directly as a limitation rather than leaving it implicit in a sentence about GOSIF's construction that a reader could easily read past without connecting it to the SIF-NDVI comparison specifically.

Entry 10

I'd reported the difference between 2015's lag (24.2 days) and 2018's lag (5.1 days) honestly since Entry 3, but on rereading my own paper I realized I'd never actually tried to explain it — just flagged it as unresolved and moved on. That's not the same thing as engaging with it.

Sitting with it properly, I don't think I can pin down the actual cause with the data I have. But I can lay out what's physically plausible: the two years had almost the same total rainfall deficit, so if deficit size alone controlled the lag, they should look more alike than they do.

Three things could explain the gap without me being able to test any of them — whether the shortfall landed early or late in the monsoon, whether NDVI's coarser 16-day sampling introduced more dating error in one year than the other, or whether farmers were growing different crops in these districts across the two years. I don't have sowing-date or crop-type records to check any of these, so I'm writing this up as three open candidate explanations rather than picking one and asserting it.

Entry 11

Going back through this project carefully, raw data included and not just the written output, I caught a discrepancy that I need to record honestly rather than quietly patch and move on from, in keeping with how the rest of this log has handled every other mistake.

What was found. `data/processed/clipped/` has held two files per date ever since Entry 4: the original rectangular-window clip (named `GOSIF_<year><doy>.tif`, no suffix — the output of the pre-fix version of `clip_gosif.py`) and the corrected polygon-mask clip (named `GOSIF_<year><doy>_clipped.tif`, from the rewritten `clip_gosif.py` that uses `rasterio.mask.mask` against the real Marathwada boundary). I never deleted the old files after the rewrite. That alone would have been harmless — except `aggregate_sif.py`'s glob pattern was `"GOSIF_*.tif"`, and its filename regex was `r"GOSIF_(\d{4})(\d{3})\.tif"`, which requires the date digits to be followed immediately by `.tif`. Every `_clipped.tif` file fails that regex (the `_clipped` sits in between) and gets silently printed as "unrecognized" and skipped. The plain, pre-fix files match the regex fine. So `aggregate_sif.py` — completely silently, no error, no warning — has been building `marathwada_sif_timeseries.csv` from the leaky-rectangle data this entire time, even after Entry 4 and Entry 5 explicitly re-ran "the full chain" and I believed the boundary fix had propagated everywhere.

Why Entry 4's own sanity check didn't catch this. Entry 4 reported yearly SIF means of 0.137 (2015), 0.151 (2018), 0.219 (2020) as a "sanity check" that the boundary fix hadn't changed the story. Those numbers are correct for the rectangle-clipped data — they're just not what I believed I was checking. The check compared old-rectangle output against itself under a different mental label, not against the actual polygon-masked output, because the aggregation step never touched the new files at all. The district-level analysis (`zonal_stats_sif.py`, used for the spatial maps and the SIF-rainfall correlation) was unaffected, because that script hardcodes the `_clipped.tif` filename directly rather than globbing for it — so the Pearson r = 0.837 / Spearman ρ = 0.857 result, and the spatial maps, were never wrong.

The fix. `aggregate_sif.py`'s glob and regex were both narrowed to `GOSIF_*_clipped.tif` / `r"GOSIF_(\d{4})(\d{3})_clipped\.tif"`, so it can only ever pick up the actual current output of `clip_gosif.py`, regardless of what stale files happen to still be sitting in the same folder. `marathwada_sif_timeseries.csv`, `marathwada_sif_ndvi_merged.csv`, `sif_ndvi_lag_by_threshold.csv`, and `final_summary_rainfall_lag.csv` were all regenerated from the properly polygon-masked rasters, and the two dependent figures (`sif_vs_ndvi_seasonal_v2.png`, `sif_ndvi_lag_by_threshold.png`) were re-rendered.

Did the finding change? Not qualitatively. Corrected mean lag: 24.3 days (2015), 6.9 days (2018), 28.7 days (2020) — versus the previously reported 24.2 / 5.1 / 25.5. The drought-year average moves from 14.6 to 15.6 days; the normal year moves from 25.5 to 28.7 days. SIF still leads NDVI's decline in all fifteen threshold-year combinations, and the drought-years-still-average-a-smaller-lag-than-the-normal-year result (the rejection of H3) still holds, now on the actually-correct boundary. The 2018 number moved the most in relative terms (5.1 → 6.9 days, about a 35% shift) — worth flagging specifically since it's the smallest, most attention-grabbing lag value in the whole study, and it moved more than the others did.

What this means more broadly. I still have the stale, un-suffixed rectangular files sitting in `data/processed/clipped/` alongside the correct ones. I'm leaving them in place for now rather than trying to delete them myself, since silently deleting data during a project review isn't the right call — but they should be removed before this repository is considered final, precisely because their continued presence is what let this bug hide for as long as it did. Any future script that globs this folder by a loose `GOSIF_*.tif` pattern will reproduce the same failure mode until they're gone.

Entry 12

Every lag value reported in this project so far, including through the Entry 11 correction, comes from a single method: threshold-crossing. That method has a specific bias built into it — it's most sensitive to *when decline starts*, particularly at the high thresholds (90%, 80%). It says nothing about whether that same conclusion would hold if I asked the question a genuinely different way. Before treating this project as finished, I wanted a second opinion from a method that isn't just a variant of the same idea.

The method. Time-lagged cross-correlation: interpolate SIF and NDVI (both normalized 0–1, restricted to the post-peak decline window) onto a shared daily grid, de-mean both series, and find the time-shift that maximizes their correlation. This asks a different question than threshold-crossing — not "when does decline start," but "what single shift best aligns the whole shape of the two curves." If both methods agree, that's real corroboration, not just the same number computed twice.

A mistake I caught before it became one. My first pass allowed lags up to 60 days, and two of the three years' peaks landed suspiciously close to that boundary (2020 peaked at lag 56 of 60). That's a classic cross-correlation artifact — testing lags close to the full series length leaves too few overlapping points to trust the estimate, and the "peak" is really just the search window running out rather than a genuine optimum. I capped the tested lag at N/4 of the decline window (a standard guideline, ~32 days here) and reran it. All three years' peaks are now safely interior to the search range, not pinned to the edge — a real optimum, not an artifact of where I stopped looking.

The result. Cross-correlation-maximizing lag: 13 days (2015), 4 days (2018), 7 days (2020) — all three positive, all at high correlation (0.99, 0.99, 0.99), and none at the search boundary. Every year's optimal lag being positive independently confirms the project's central finding: SIF leads NDVI's decline, in all three years, by a wholly different method than the one used everywhere else in this study. That's a real strengthening of H1 specifically.

What it does *not* confirm, and I'm not going to pretend otherwise. The threshold-crossing method ranked the three years' lag magnitude as 2020 (28.7d) > 2015 (24.3d) > 2018 (6.9d). Cross-correlation ranks them 2015 (13d) > 2020 (7d) > 2018 (4d). Both methods agree 2018 has the smallest lag — that part holds up twice over. But they disagree on whether 2015 or 2020 has the largest lag, which is exactly the comparison underlying the H3 (drought-amplifies-lag) discussion. I thought about only reporting the parts that agree, but that would be quietly cherry-picking a cleaner story than the data actually supports. The honest version is: the *qualitative* finding (SIF leads NDVI) is now more robust than before, because two different methods agree on it. The *specific numbers*, and the *exact ranking* between years, are more uncertain than a single-method analysis would suggest, because two reasonable methods don't agree on them. Both of those statements are now in the paper, in the dashboard, and here.

Where this landed. New script `src/analysis/cross_correlation_lag.py`, new output `data/processed/cross_correlation_lag_by_year.csv`, new figure `cross_correlation_lag.png`, new GA_Research_Paper.md Section 3.6 (methodology) and 4.6 (results), an added caveat sentence in the Discussion and a new Limitations bullet, a new Lag Analysis dashboard section, and a seventh Key Finding in GA_Project_Report.md. Nothing about the original threshold-crossing numbers changed — this is purely additive, a second lens on the same question rather than a replacement for the first one.

Entry 13

Once the paper, journal, dev log, and dashboard were all in a stable state, I sat down and went back through the whole project critically, the way I'd want a thesis committee to, rather than treating "it runs and the numbers look right" as the finish line. Going back through with fresh eyes turned up the same list of gaps every time: no uncertainty quantification around the lag estimates, the district-level correlation's non-independence being asserted but never measured, the Google Earth Engine acquisition step still being unscripted, and — the one that bothered me most — the observation that RQ3, as originally posed in my own planning documents, had never actually been answered in the final paper. This entry records what I did about each of those, and what I tried and abandoned along the way.

1. Bootstrapping the cross-correlation lag estimates — first attempt failed, and why. My first instinct was a moving-block bootstrap directly on the interpolated daily SIF/NDVI series: chop each year's decline curve into overlapping 10-day blocks, resample the blocks with replacement, recompute the lag on the shuffled reconstruction, repeat 2,000 times. Every single year's resulting confidence interval collapsed to `[0.0, 0.0]` — not a tight interval, a *degenerate* one. That result would have been actively misleading if I'd reported it as "the lag is 0 days ± nothing." What's actually happening: these decline curves are strongly non-stationary (they trend monotonically downward across the season), and shuffling blocks of a trending curve destroys the trend itself — the resampled series stops looking like a decline curve at all, so the cross-correlation procedure just finds whatever's left, which is noise centered near zero lag. I was measuring "what happens when you erase the signal," not sampling uncertainty in the signal.

I replaced it with case-resampling: instead of shuffling points on the interpolated daily grid, resample *which of the original 17 sparse 8-day satellite observations* went into building that year's curve, with replacement, then re-interpolate and re-run the identical cross-correlation procedure from Entry 12. This preserves the actual shape of the decline curve in every replicate while still capturing genuine uncertainty — specifically, the uncertainty from not knowing in advance which cloud-affected overpass dates you'll actually get in a given season. Results this time were sensible: 2015, 13 days [95% CI 5–20]; 2018, 4 days [0–9]; 2020, 7 days [0–32]. Every pairwise comparison between years overlaps — meaning the exact ranking discussion in Entry 12 is, honestly, not something the data can support at high confidence, even though the *direction* is: the lag was non-negative in 100% of bootstrap replicates for every single year, and strictly positive in 84.5–99.9% of them. That's a real, quantified strengthening of H1 and a real, quantified weakening of confidence in the specific numbers — both true at once, and both now stated as such rather than left as a single point estimate that looked more precise than it was.

2. Quantifying the district-level correlation's non-independence with Moran's I. GA_Research_Paper.md has said, since it was first written, that the 24 district-year observations behind the $r = 0.837$ correlation aren't fully independent because the eight districts are geographically adjacent. That was always true, but it was an assertion, not a measurement. I built a Queen-contiguity spatial weights matrix from the actual FAO GAUL district polygons already used throughout this project (mean 3.25 neighbors per district) and ran Moran's I, with a 9,999-permutation significance test, for both mean SIF and rainfall anomaly, separately for each study year. All six year × variable combinations came back significant ($p < 0.05$), with I ranging from 0.26 to 0.55 against an expected value of −0.14 under spatial randomness. This doesn't change the correlation number or invalidate it as evidence — the underlying spatial pattern is real, and it's independently cross-validated against the SIF maps in Section 4.3 — but it does mean I can now say precisely, instead of just gesturing at it, that the effective sample size is closer to 3 than 24.

3. Scripting the Earth Engine acquisition step — with an honest caveat about what "scripting" does and doesn't mean here. The NDVI, cropland-mask, and CHIRPS extractions were run by hand in Earth Engine's Code Editor from the start of this project, and every version of the Limitations section has said so. I wrote `src/acquisition/gee_data_acquisition.py`, a Python translation using `earthengine-api` and `geemap` that reconstructs exactly what Section 3.1 of the paper describes — SummaryQA-filtered MOD13Q1, the MCD12Q1 cropland mask (classes 12 and 14), CHIRPS seasonal totals and the 20-year climatology, all against the FAO GAUL Marathwada polygon. I want to be precise about what this does and doesn't fix: I do not have an authenticated Earth Engine account in this environment, so I have not run this script and cannot confirm it reproduces the original interactive output exactly. What it does do is turn a purely prose-described, only-mentally-reproducible step into a checked-in, version-controlled artifact that anyone with EE access can actually run and inspect. I've stated this distinction explicitly in the Limitations section rather than letting the mere existence of the file imply more than it's earned.

4. Actually answering RQ3. This is the one I'm least comfortable having missed originally. RQ3 and the research-questions list in GA_Project_Report.md always framed the policy motivation partly in terms of the *official* drought declaration timeline, not just rainfall deficit — but GA_Research_Paper.md's Section 4 never once compared SIF or NDVI against an actual declaration date. Section 3.5's rainfall-deficit comparison answered a real question, just not that one. I searched for a verifiable official Maharashtra drought-declaration date for each study year. For 2018 this was straightforward and well-documented: the state government declared drought in 151 talukas across 26 districts, including all eight Marathwada study districts, on

31 October 2018 (Zee News, 2018; EPW, 2018). For 2015 I did not find a comparably clean single date — coverage of the 2015 Marathwada drought is dominated by the subsequent 2016 water-crisis reporting rather than a dated 2015 Kharif declaration — so I'm reporting the comparison for 2018 only and saying so explicitly, rather than picking a plausible-sounding date I couldn't actually verify.

The 2018 comparison itself: SIF crossed its 90%-decline threshold on 8 September 2018, NDVI on 12 September 2018, the official declaration came on 31 October 2018 — 53 and 49 days later respectively. Sitting with that number changed how I think this study's own policy argument should be framed. The SIF-vs-NDVI gap that's the whole point of this paper is 3-4 days. The satellite-vs-official-declaration gap is roughly seven weeks. Those aren't the same order of magnitude, and reporting the RQ3 comparison honestly means saying so: the bigger, more actionable finding here isn't "SIF beats NDVI by a few days," it's "any satellite-based signal beats the current 7-week-plus institutional process by a lot, and SIF is a specific, physiologically-motivated way to make that already-large improvement slightly larger." I'd rather state the policy case at the size it's actually earned than let the paper's narrower technical finding imply a bigger one.

Where this all landed. New scripts: `src/analysis/bootstrap_lag_uncertainty.py`, `src/analysis/spatial_autocorrelation.py`, `src/analysis/rq3_official_declaration_comparison.py`, `src/acquisition/gee_data_acquisition.py`. New data outputs: `cross_correlation_lag_bootstrap_ci.csv`, `spatial_autocorrelation_morans_i.csv`, `rq3_official_declaration_comparison.csv`, `official_drought_declarations.csv`. New figures: `cross_correlation_lag_bootstrap_ci.png`, `spatial_autocorrelation_morans_i.png`, `rq3_declaration_timeline_2018.png`. GA_Research_Paper.md gained three new methodology subsections (3.7–3.9), three new results subsections (4.7–4.9), an expanded Discussion, four revised/new Limitations bullets, a revised Conclusion, and three new references. GA_Project_Report.md gained a new Section 5.8, three new QA entries, three new Key Findings (8, 9, 10), and two new Limitations bullets. The dashboard, README, and PORTFOLIO_STATUS.md were updated to match. Nothing about the core findings from Entries 1–12 changed — every addition in this entry is either a precision improvement (bootstrap CI, Moran's I) or a genuinely new comparison this project should have run from the start (RQ3) — but several claims are now stated with more appropriate, and in places more modest, confidence than before.

Entry 14

Every version of this paper's Limitations section has said the same thing: three years (two drought, one normal) is a small sample, and the two drought years differ substantially from each other. That's the single limitation I keep coming back to as the one actually worth fixing rather than just stating. Decided to add five more years — 2016, 2017, 2019, 2022, 2023 — bringing this study to eight years total.

First step: getting the raw data. GOSIF v2's 8-day product covers 2000–2024, so all five new years exist. Wrote `src/acquisition/download_gosif_new_years.py` to pull them in the same DOY-153-to-361 seasonal window already used for 2015/2018/2020, same file-naming convention, so `clip_gosif.py` needs no changes to pick the new files up. Also extended `src/acquisition/gee_data_acquisition.py`'s `STUDY_YEARS` list to all eight years, and added two export blocks it didn't have before (region-mean rainfall and by-district rainfall as proper CSVs) — the original script only printed rainfall totals to the console, which was fine for copying three numbers by hand but not eight.

One genuine surprise while setting this up: I can't reach the GOSIF data server directly from this environment (403 Forbidden on every request, browser user-agent or not) — the same kind of IP-range block that stopped Sentinel Hub in DOUBLE_JEOPARDY. So the download has to run on my own machine, same as that project's satellite pulls did.

Not yet done, and I want to be upfront about it: neither script has actually been run yet. The GOSIF download needs to happen from a normal home connection, and the GEE extraction needs my authenticated Earth Engine account, same requirement the original (unrun) acquisition script already had. Every number in this project as of this entry is still the original 3-year result. This entry marks the start of the expansion, not its completion — the next entry will cover what actually came back once both scripts have run and I've re-processed the full pipeline (clipping, lag analysis, cross-correlation, bootstrap CIs, Moran's I, and every downstream document and dashboard page) against eight years instead of three.

Entry 15

Ran `download_gosif_new_years.py` first. All 135 files (5 years × 27 composites) came down clean on the first attempt — no repeat of the GOSIF-blocking problem, since this ran from my own connection as expected.

`gee_data_acquisition.py` was a different story — every one of the next three problems was a genuine bug in code that had never actually been executed before, not a repeat of anything already tested.

Bug 1 — no registered Cloud project. Modern Earth Engine now requires `ee.Initialize(project=...)` with a registered Google Cloud project; the script only had a bare `ee.Initialize()` and a commented-out `ee.Authenticate()`, both leftovers from before this requirement existed. Added both properly, using my own EE project ID.

Bug 2 — MODIS clip against the wrong projection. The NDVI extraction failed with a projection-transform error trying to clip a MOD13Q1 image (native sinusoidal grid, SR-ORG:6974) against the Marathwada boundary polygon (EPSG:4326) — some district boundary geometries just don't transform cleanly onto that edge. The `.clip()` call turned out to be redundant anyway, since the `reduceRegion()` call right after it already restricts the mean calculation to the same geometry. Removed it.

Bug 3 — a whole district silently missing. After the above two fixes, the run completed without errors, but I only trusted that on faith for about five minutes before actually counting rows in the output. The by-district rainfall export had 56 rows, not the 64 (8 districts × 8 years) it should. The `MARATHWADA_DISTRICTS` list used the spelling `"Beed"` — correct in common English usage, but not what FAO GAUL 2015's `ADM2_NAME` field actually contains for that district, which is `"Bid"`. `ee.Filter.inList()` doesn't error on a spelling that matches nothing; it just silently drops that district from the merged region boundary and every district-level export, for every one of the eight years, since the region-level boundary is built from the same district list. Checked the correct spelling against the district GeoJSON already sitting in `data/raw/` and fixed the list. Re-ran the whole extraction a second time after that fix, then actually opened the output CSV and counted: 8 districts × 8 years = 64 rows, all eight names present. That's the version everything downstream now uses.

What actually came back, once the data was trustworthy. Ran `rainfall_analysis.py` against the corrected 8-year regional rainfall totals, and this produced the most interesting result of this whole expansion pass, one I hadn't expected going in. Using the same drought threshold implied by the original study (roughly half a climatological standard deviation below the 2001–2020 mean), 2015 (−21.5%, z = −1.12) and 2018 (−18.3%, z = −0.95) are still the two driest years — same as the original 3-year study called them. But none of the five newly added years come close to that threshold: 2016 (+9.5%), 2017 (−1.1%), 2019 (+13.3%), 2022 (+36.8%), 2023 (−3.7%) are all within roughly one standard deviation of normal. So across the full 8-year record, only 2 of 8 years (25%) actually qualify as drought years by this study's own definition — the original 3-year sample (2 of 3, 67%) was, by pure chance of which years got picked first, far more drought-heavy than the region's real 8-year climate record turns out to be. That's worth stating plainly in the paper's limitations discussion: it doesn't undermine the drought-year findings themselves, but it does mean the *proportion* of drought years in the original sample wasn't representative, and the 8-year record is a more honest picture of how often Marathwada actually sees a season this dry.

I also switched every hardcoded `DROUGHT_YEARS = {2015, 2018}` across the lag, cross-correlation, and bootstrap scripts to derive that set from the actual rainfall anomaly output instead of a literal — it still resolves to the same two years right now, but it means the classification is computed from data going forward instead of typed in by hand, which matters more with 8 years in play than it did with 3.

Still pending as of this entry: `clip_gosif.py` and `aggregate_sif.py` — the two scripts that turn the raw GOSIF rasters into the actual SIF time series — need to run against all 216 raw files (27 composites × 8 years), and that has to happen on my own machine rather than remotely, purely because 11+ GB of raw rasters isn't worth moving anywhere; the clipped output per file is under 40 KB, so only that tiny output needs to travel anywhere else. Once that's run, the SIF-side findings (lag analysis, cross-correlation, bootstrap CIs, Moran's I, district-level correlation) all get reprocessed against the full 8 years — those are still the original 3-year numbers as of this entry.

Entry 16

Ran `clip_gosif.py` and `zonal_stats_sif.py` against all 216 raw GOSIF files on my own machine, same as the acquisition scripts in Entry 15. Both are fully generic — glob-based file discovery, no hardcoded year list anywhere in either script — so neither needed a single code change to pick up the five new years. That was a deliberate design choice back when I originally wrote them for three years, and it paid off exactly as intended here. `marathwada_sif_timeseries.csv` came back with 216 rows (27 composites × 8 years) and `sif_by_district.csv` with 64 rows (8 districts × 8 years), matching the region-level rainfall row counts from Entry 15 exactly — a useful cross-check that nothing silently dropped a year or a district on the SIF side either.

With real 8-year SIF data in hand, I reprocessed every downstream analysis script against it: `compare_sif_ndvi.py`, `lag_analysis.py`, `cross_correlation_lag.py`, `bootstrap_lag_uncertainty.py`, `sif_rainfall_correlation.py`, `spatial_autocorrelation.py`, and `rq3_official_declaration_comparison.py`. Also switched every remaining hardcoded `DROUGHT_YEARS = {2015, 2018}` literal to the dynamic, rainfall-anomaly-derived version already started in Entry 15, so the classification is computed consistently everywhere rather than typed in by hand in some scripts and derived in others. Two smaller deprecation warnings surfaced along the way and got fixed while I was already in those files: `matplotlib.cm.get_cmap` (replaced with `matplotlib.colormaps[...].resampled(...)`) and `geopandas`'s `.unary_union` (replaced with `.union_all()`).

The 2018 exception. This is the single most interesting thing to come out of reprocessing. Every lag-estimation method — threshold-crossing, cross-correlation, and the bootstrap check on top of that — agrees that 2018 no longer looks like a normal drought year in this study's own central finding. Its mean threshold-crossing lag is marginally negative (−1.1 days, against 4.2–23.8 days for every other year), its cross-correlation lag is exactly zero, and only 8.1% of its bootstrap replicates show a positive lag versus 42–97% for every other year. The RQ3 comparison against the official 2018 drought declaration confirms it a fourth way: NDVI now crosses its 90%-decline threshold *before* SIF that year, the reverse of what the original three-year study reported. I don't have crop-type, sowing-date, or any other covariate data that would explain why 2018 specifically differs, so I'm reporting it as a genuine, unexplained finding rather than guessing at a cause or quietly averaging it into the drought-year group.

The other headline number: only 2 of 8 years are actually drought years. I flagged this in Entry 15 from the rainfall side alone; it holds up identically once SIF is in the picture too. The original three-year sample being two-thirds drought years was, in hindsight, a considerably more drought-heavy selection than Marathwada's actual eight-year climate record supports.

Weaker, but still real, at the larger sample. The district-level SIF-rainfall correlation dropped from r = 0.837 (n = 24) to r = 0.567 (n = 64) — still highly significant, but a genuine weakening, not a rounding difference, and I'm reporting it as exactly that rather than emphasizing the still-significant p-value and burying the drop. Moran's I tells a similarly two-sided story: rainfall's spatial clustering stayed significant in all 8 years, same as it was 3-for-3 originally, but SIF's spatial clustering is now significant in only 4 of 8 years — real in some years, not detectably present in others, which is a more qualified picture than the original "all three years significant" result implied.

Updating everything downstream, properly, not just the numbers that were easy to find. Once the reprocessed CSVs and figures were in hand, I went through every dashboard page, both PDFs, the README, the citation metadata, and the policy brief one at a time rather than doing a global find-and-replace on the old numbers — a few pages needed more than a number swap. The district-level SIF ranking needed an actual rewrite, not just new numbers: Nanded was the highest-SIF district in all three original years, but at eight years it's highest in only 2 of 8, with Aurangabad leading most often (4 of 8) and two other districts leading once each — the "consistently highest" framing from the original study simply doesn't hold at this sample size, so I said so directly instead of picking whichever district happened to average highest and calling it consistent. The bottom of that ranking is much more stable: Osmanabad is lowest in 7 of 8 years, with Bid taking over specifically in 2018 — the same district that recorded the single lowest SIF value of the entire study, and also the year's biggest rainfall deficit. The Lag Analysis dashboard page needed the most rework, since its cross-correlation and bootstrap sections had never been updated past the original three-column layout — I rebuilt both as full eight-year metric grids with corrected interpretive text, since the original "every year's lag is strictly positive" claim is no longer true at eight years (four years now tie at exactly zero).

A repository cleanup, while I was touching every document anyway. I found three orphaned files sitting in the repository root — `Development_Log.md`, `Project_Journal.md`, and `Research_Paper.md` — leftover, unlinked duplicates of the current `GA_`-prefixed documents from before I adopted that naming convention. They still had the original three-year numbers and weren't referenced anywhere except in a handful of stale in-text citations inside the current documents (pointing to entry numbers and section numbers in the old files by their old bare names). Nobody browsing this repository should be able to open a "Research Paper" and a "GA Research Paper" side by side and find two different correlation coefficients for the same study. I deleted all three orphaned files and repointed every internal reference — in the paper, the report, this log, the dashboard, and the acquisition/analysis script comments — to the current `GA_`-prefixed filenames.

Regenerating the PDFs. `GA_Research_Paper.pdf` and `GA_Project_Report.pdf` didn't exist yet as files — only their `.md` sources did — so I built both fresh from the updated markdown via pandoc and wkhtmltopdf, and rebuilt `Policy_Brief.pdf` from its rewritten source too, since the copy on disk still had the original three-year figures. All three were checked page-by-page by rendering them to images before calling any of this done; the research paper now runs 22 pages with all 16 figures embedded correctly (an earlier attempt lost the image paths because of a relative-path issue in the HTML build step, caught by the same page-by-page check before it went anywhere).

Where this landed. Every dashboard page (1–10), `app.py`, the README, `CITATION.cff`, `Policy_Brief.md`, `GA_Research_Paper.md`, and `GA_Project_Report.md` are now consistent with each other and with the actual eight-year data on disk — same correlation coefficients, same drought-year count, same 2018 caveat, same district rankings, everywhere they're mentioned. This entry closes out the sample-size expansion that Entry 14 started: every number in this project, across every document and every dashboard page, is now the real eight-year result, not a partially-updated mix of old and new.